

# Input Content (IC)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Empirical dataset inspection partially resolves what the paper does not document. Ethiopian calendar months (Meskerem, Hamle) appear as DATE slot values [\[DATASET-D1, D2, D6\]](#), Ethiopian intercity cities (Dire Dawa, Bahir Dar, Hawassa, Arba Minch) appear as CITY\_OR\_PROVINCE slots [\[DATASET-D2, D9\]](#), and Ethiopia-specific financial entities (Amhara Bank, Commercial Bank of Ethiopia, TeleBirr, Dashen Bank) appear in banking-domain instances [\[DATASET-D3, D4, D5, D34\]](#). This is materially better than the registry feared. However, several deployment-critical content gaps remain: (a) Ethiopian Birr (ETB) was absent as a CURRENCY/MONEY slot value in any sampled Amharic example despite Naira/Rand appearing in other tracks [\[DATASET-D5, D19, D22 vs Amharic absence\]](#); (b) most Amharic DATE slots used relative expressions rather than named Ge'ez months (only 2/9) [\[DATASET-D15, D16, D28\]](#); (c) Ethiopian time convention (6-hour offset) is unaddressed in TIME annotations [\[DATASET-D11, D23\]](#); (d) no code-switched Amharic–Oromo / Amharic–Tigrinya utterances appear, despite ~70% L2 Amharic users in the deployment population [\[DATASET-D1–D18 monolingual\]](#). IC was marked HIGH priority in elicitation, so these residual gaps weigh heavily.

| ID  | Evidence                                                                                                                                                                                                                                                                  |
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| D1  | ከድረዳዋ ወደ ባህርዳር ለመስከረም እምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."                                                                                                                                                                 |
| D2  | target: "CITY_OR_PROVINCE: ድረዳዋ \$\$\$ CITY_OR_PROVINCE: ባህርዳር \$\$\$ DATE: መስከረም እምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)                                                                           |
| D3  | የአማራ ባንክ አካውንቱ ስንት አለው? / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated                                                                                                                                          |
| D4  | ንግድ ባንክ ላይ የቀረኝ ገንዘብ አለ? / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank                                                                                                                                   |
| D5  | በቴሌብር የብድር አካውንቴ ላይ ... / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform                                                                                                                   |
| D6  | ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ? / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar                                                                                                         |
| D7  | ወደ ቱርክ በፍጥነት የምድርስበት የበረራ አማራጭ ንገረኝ? / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking                                                                                            |
| D8  | ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus                                                                                                                                      |
| D9  | raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding                                                             |
| D10 | raw: "ከቴሌብር አካውንቴ ገንዘብ ለማን ለማን እንደላኩ ዝርዝሩን አሳየኝ?" / text: "Show me history of my telebirr transfers?" — TeleBirr payment company referenced in Amharic utterance                                                                                                          |
| D11 | ነገ ጠዋት 12 ሰአት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰአት"                                                                                                        |
| D12 | Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiyya irraa naaf kuti — "Book me an Ethiopian Airlines flight from Dire Dawa to Bole then to Jimma." — spans empty despite city names and airline                                                 |
| D13 | Ebila 12 hanga Ebila 14tti Mattuu keessatti kireeffannaa konkolaataa argachuun ni danda'amaa / target: "DATE: Ebila 12 hanga Ebila 14 \$\$\$ CITY_OR_PROVINCE: Mattuu — Oromo month name "Ebila" (April) used as date slot — culturally grounded Ethiopian Oromo calendar |

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| D14 | Sagantaa sirna cidhaaf sadaasa 21tti qabame kalaandarii koo irraa haqi maloo" / target: "CALENDAR_EVENT: sirna cidhaa \$\$ DATE: sadaasa 21 — Oromo month "Sadaasa" (November in Oromo calendar) used as date slot          |
| D15 | የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ::" / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name                                                        |
| D16 | ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መከራየት እፈልጋለሁ::" / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month                                   |
| D17 | በቶቶት ለቃለአብ ቦታ ማስያዝ እንዳለብኝ..." / target: "RESTAURANT_NAME: ቶቶት \$\$ PERSONAL_NAME: ቃለአብ — Restaurant "ቶቶት" (Totot) — Ethiopian cultural reference; personal name "ቃለአብ" is distinctly Ethiopian                              |
| D18 | የቦሎ ማሳደሻ ስንት ነው" / target: "BILL_TYPE: የቦሎ ማሳደሻ — "What is the Bole maintenance/upgrade fee?" — "ቦሎ" likely refers to Bole (Addis Ababa district)                                                                           |
| D19 | Naira nawa gare ni ... na bankin Gt?" / target: "CURRENCY: Naira \$\$ BANK_NAME: Gt — Nigerian Naira currency annotated in Hausa — confirms per-language currency grounding                                                 |
| D22 | Ndinganayo imali engange R700 kwiakhawunti yam yaseStandard Bank?" / target: "MONEY: R700 \$\$ BANK_NAME: yaseStandard Bank — South African Rand (R700) and Standard Bank in Xhosa                                          |
| D23 | "ነገ ጠዋት 12 ሰአት..." (no span annotation) — "12 ሰአት" (12 o'clock) unannotated — "ጠዋት" (morning) absent from target; Ethiopian time convention unclear                                                                         |
| D28 | በዚህ ሳምንት ምንም ቀጠሮ እንዲኖረኝ አልፈልግም::" / target: "DATE: በዚህ ሳምንት — Relative week reference, no Ethiopian calendar month                                                                                                          |
| D34 | raw: "እባካችሁ የዳሽን ባንክ ሂሳቤን ታረጋግጡልኛለችሁ?" / text: "can you please check my dasion bank account" — "Dashen Bank" (major Ethiopian bank) — Amharic source is culturally grounded but English transliteration is lossy ("dasion") |

## Strengths

- Ethiopian calendar months (Meskerem, Hamle) confirmed as valid DATE slot annotations in travel-domain Amharic instances [DATASET-D1, D2, D6]
- Major Ethiopian intercity cities (Dire Dawa, Bahir Dar, Hawassa, Arba Minch) appear as CITY\_OR\_PROVINCE slots [DATASET-D2, D9]
- Ground-up cultural elicitation methodology rather than translation [Q6, Q13, Q30]
- Ethiopia-specific banks and TeleBirr appear as natively grounded entities [DATASET-D3, D4, D5]

| ID  | Evidence                                                                                                                                                                                                                                                                           |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q6  | "The data construction process starts with providing an annotator with sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, and they are to come up with culturally-relevant similar sentences and relevant slot entities (see Figure 1)." (p.1)  |
| Q13 | "However, these efforts face two key challenges: (1) the translationese effect, which makes utterances sound less natural in the target languages (Vanmassenhove et al., 2021; Bizzoni et al., 2020), and (2) the creation of utterances that are less culturally relevant." (p.1) |
| Q30 | "In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)                    |
| D1  | ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ:: — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."                                                                                                                                                                         |
| D2  | target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$ CITY_OR_PROVINCE: ባህርዳር \$\$ DATE: መስከረም አምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)                                                                                        |
| D3  | የአማራ ባንክ አካውንቱ ስንት አለው" / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated                                                                                                                                                   |
| D4  | ንግድ ባንክ ላይ የቀረኝ ንገዘብ አለ" / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank                                                                                                                                            |
| D5  | በቴሌብር የብድር አካውንቱ ላይ ..." / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform                                                                                                                           |
| D6  | ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar                                                                                                                 |
| D9  | raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding                                                                      |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                                        |
|------|----------------------------------------------------------------------------|
| IC-1 | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| IC-2 | Does culturally sensitive content align with target culture?               |
| IC-3 | Flag inputs assuming Western-specific knowledge                            |
| IC-4 | Regional annotator recruitment for culturally sensitive instances          |
| IC-5 | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Yes — deployment demands Amharic city names/abbreviations, Ethiopian calendar dates, Birr fares, and operator-specific terminology, confirmed by elicitation Q2 (HIGH priority) and YAML calendar/toponym sections.

**IC-2:** Partially. Cultural grounding is confirmed for banking-domain Amharic [DATASET-D3, D4, D5] and partially for travel (Ethiopian cities + Meskerem present) [DATASET-D1, D2, D9]. The grounding methodology is documented [Q30, Q40, Q41].

**IC-3:** Western-content leakage was not observed in the Amharic sample (though present in some other tracks, e.g., Hausa Taco Bell [DATASET-D31]). Air-travel-to-international-destinations framing dominates the Amharic travel domain [DATASET-D7, D8].

**IC-4:** Native Amharic speakers annotated [Q7], but regional/L2 demographics undisclosed and code-switched utterances absent from sample [DATASET-D1–D18]; full re-annotation by Ethiopian regional pool would be advisable.

**IC-5:** Documented residual content gaps: (a) Birr/ETB absent from sampled Amharic CURRENCY/MONEY slots; (b) Ethiopian calendar months sparse vs. relative dates [DATASET-D15, D16, D28]; (c) Ethiopian clock convention undocumented; (d) no code-switched utterances [DATASET-D1–D18].

| ID  | Evidence                                                                                                                                                                                                                                                                          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q2  | "Intent detection and slot-filling are crucial components of the natural language understanding module in task-oriented dialogue systems (Hemphill et al., 1990; Coucke et al., 2018; Gupta et al., 2018)." (p.1)                                                                 |
| Q6  | "The data construction process starts with providing an annotator with sentences from the CLINC dataset (Larson et al., 2019a) with a specified intent type, and they are to come up with culturally-relevant similar sentences and relevant slot entities (see Figure 1)." (p.1) |
| Q7  | "utterances generated by native speakers across diverse domains, including banking, travel, home, and dining." (p.1)                                                                                                                                                              |
| Q30 | "In our paper, we introduce INJONGO which is the largest intent detection and slot-filling dataset covering 16 African languages, and we ensured that the slot entities are more culturally relevant in the respective countries the languages are from." (p.2)                   |
| Q40 | "A Xhosa annotator was asked to generate another utterance belonging to the same intent type but capturing the South African context where the language is spoken." (p.3)                                                                                                         |
| Q41 | "Thus, the annotator used the R200 as "money" with currency Rand (R), and more familiar South African banks such as "FNB" and "Absa" for "bank name" slot." (p.3)                                                                                                                 |
| D1  | ከድሬዳዋ ወደ ባህርዳር ለመስከረም እምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."                                                                                                                                                                         |
| D2  | target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$ CITY_OR_PROVINCE: ባህርዳር \$\$ DATE: መስከረም እምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)                                                                                       |
| D3  | የአማራ ባንክ አካውንቱ ስንት አለው" / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated                                                                                                                                                  |
| D4  | ንግድ ባንክ ላይ የቀረጽ ገንዘብ አለ" / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank                                                                                                                                           |
| D5  | በቴሌብር የብድር አካውንቱ ላይ ..." / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform                                                                                                                          |
| D6  | ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ?" / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar                                                                                                                |

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| D7  | ወደ ቱርክ በፍጥነት የምደርስበት የበረራ አማራጭ ንገረኝ። / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking                                                                                            |
| D8  | ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus                                                                                                                                      |
| D9  | raw: "ለቀጣይ የጉብኝት ጉዞ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding                                                              |
| D10 | raw: "ከቴሌ ብር አካፊንቴ ገንዘብ ለማን እንደላኩ ዝርዝርን አሳየኝ?" / text: "Show me history of my telebirr transfers?" — TeleBirr payment company referenced in Amharic utterance                                                                                                             |
| D11 | ነገ ጠዋት 12 ሰአት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰአት"                                                                                                        |
| D12 | Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiya irraa naaf kuti — "Book me an Ethiopian Airlines flight from Dire Dawa to Bole then to Jimma." — spans empty despite city names and airline                                                  |
| D13 | Ebila 12 hanga Ebila 14tti Mattuu keessatti kireeffannaa konkolaataa argachuun ni danda'amaa / target: "DATE: Ebila 12 hanga Ebila 14 \$\$\$ CITY_OR_PROVINCE: Mattuu — Oromo month name "Ebila" (April) used as date slot — culturally grounded Ethiopian Oromo calendar |
| D14 | Sagantaa sirna cidhaaf sadaasa 21tti qabame kalaandarii koo irraa haqi maloo / target: "CALENDAR_EVENT: sirna cidhaa \$\$\$ DATE: sadaasa 21 — Oromo month "Sadaasa" (November in Oromo calendar) used as date slot                                                       |
| D15 | የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ። / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name                                                                                                        |
| D16 | ከሚመጣው ወር ጀምሮ ለ3 ወር ትየታ መኪና መከራየት እፈልጋለሁ። / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month                                                                                   |
| D17 | በቶቶ ለቃለኡብ ቦታ ማስያዝ እንዳለብኝ..." / target: "RESTAURANT_NAME: ቶቶ \$\$\$ PERSONAL_NAME: ቃለኡብ — Restaurant "ቶቶ" (Totot) — Ethiopian cultural reference; personal name "ቃለኡብ" is distinctly Ethiopian                                                                             |
| D18 | የቦሎ ማሳደሻ ስንት ነው / target: "BILL_TYPE: የቦሎ ማሳደሻ — "What is the Bole maintenance/upgrade fee?" — "ቦሎ" likely refers to Bole (Addis Ababa district)                                                                                                                          |
| D28 | በዚህ ሳምንት ምንም ቀጠሮ እንዲኖረኝ አልፈልግም። / target: "DATE: በዚህ ሳምንት — Relative week reference, no Ethiopian calendar month                                                                                                                                                          |
| D31 | Za ku iya tabbatar min ina da ajiyar tebur a taco bell da karfe 7 na yamma / target: "RESTAURANT_NAME: taco bell — "Taco Bell" — US fast-food chain, Western entity in Hausa utterance                                                                                    |

Information Gaps

- Birr coverage in the 99.4% of Amharic data not sampled
- Frequency of named Ethiopian calendar months vs. relative dates in full travel split
- Whether any code-switched Amharic utterances exist in unsampled portions

Requires Expert Verification

- Native Amharic-speaking expert review of Ethiopian time convention handling in TIME slots across the full dataset

Remediation

**Gap 1: Ethiopian Birr (ETB) absent as CURRENCY/MONEY slot in sampled Amharic; most DATE slots use relative expressions rather than named Ge'ez months; Ethiopian clock convention undocumented; no code-switched utterances**

**Recommendation:** Construct a 'ETH-Travel-Slot Stress Set' of Amharic utterances densely populated with (a) Birr fare amounts (VIP/economy tiers), (b) Ge'ez calendar month names across all 13 months including Pagume, (c) Ethiopian-clock TIME values with explicit annotator convention, (d) code-switched Amharic–Oromo and Amharic–Tigrinya tokens for cities and booking terms; evaluate alongside INJONGO Amharic test.

**Gap 2: Annotation gaps observed empirically (empty spans on temporally explicit utterances)**

**Recommendation:** Run a targeted audit pass over Amharic train/dev/test for empty-span utterances containing temporal lexemes ('ሰአት', 'ጠዋት', 'ነገ', Ge'ez month names) and patch with corrected annotations before using F1 as a model-selection metric.